Project Proposal

Scalable Graph Partitioning for Communication-Efficient Distributed Optimization

Team Members: Name 1, Name 2, Name 3

CS240: Algorithm Design and Analysis, Spring 2026

1 Topic Selection

We choose **Topic 2: Community Detection / Graph Partitioning** from the reference list, but use a different modern application: **communication-efficient distributed optimization**. In distributed optimization, a large-scale problem is solved by multiple agents or workers that exchange information through a communication graph. The graph structure directly affects communication cost, load balance, and convergence behavior. Instead of a standard social-network community detection setting, we study how classical graph partitioning algorithms can organize large-scale distributed computation. This application is closely related to large-scale optimization, multi-agent systems, and distributed machine learning.

2 Problem Formulation

Let $G = (V, E, w)$ be an undirected weighted graph, where each vertex represents a data block, variable block, agent, or computational task, and each edge represents communication or dependency between two vertices. Given an integer k, the goal is to partition V into k disjoint subsets $V_1, \ldots, V_k$.

The optimization objective is

$$\min_{V_1,\ldots,V_k} \sum_{(u,v)\in E,\ u\in V_i,\ v\in V_j,\ i\neq j} w_{uv},$$

subject to a balance constraint

$$|V_i| \leq (1+\epsilon)\frac{|V|}{k}, \quad i = 1, \ldots, k.$$

Here the cut objective models cross-partition communication cost, and the balance constraint models computational load balance. The input is a weighted graph and the desired number of partitions; the output is a balanced k-partition of the vertices.

3 Related Work

Graph partitioning is a classical algorithmic problem in scientific computing, parallel computation, and large-scale graph processing. Spectral partitioning uses eigenvectors of the graph Laplacian to find balanced cuts. The Kernighan–Lin algorithm is a classical local-search method that iteratively swaps vertices to reduce cut size. Multilevel methods such as METIS improve scalability by graph coarsening, partitioning, and refinement. Distributed optimization over networks is closely related: in consensus and distributed gradient methods, agents update local variables while communicating with neighbors, so topology affects both convergence and communication overhead.

We plan to use the **Kernighan–Lin graph partitioning heuristic** as the specific paper/method selected for reproduction. Spectral bisection and a greedy balanced partitioning method will be used as baselines or variants. Related papers include:

- B. W. Kernighan and S. Lin, "An Efficient Heuristic Procedure for Partitioning Graphs," *Bell System Technical Journal*, 1970. DOI: 10.1002/j.1538-7305.1970.tb01770.x.

- G. Karypis and V. Kumar, "A Fast and High Quality Multilevel Scheme for Partitioning Irregular Graphs," *SIAM Journal on Scientific Computing*, 1998. DOI: 10.1137/S1064827595287997.
- A. Nedic and A. Ozdaglar, "Distributed Subgradient Methods for Multi-Agent Optimization," *IEEE Transactions on Automatic Control*, 2009. DOI: 10.1109/TAC.2008.2009515.

4 Algorithm and Technical Plan

We plan to implement and compare the following graph partitioning methods:

1. **Greedy balanced partitioning baseline.** Vertices are assigned to partitions one by one while maintaining balance and reducing local cut cost.
2. **Spectral bisection.** We compute the graph Laplacian, use the Fiedler vector to bisect the graph, and recursively apply bisection to obtain k partitions.
3. **Kernighan–Lin refinement.** Starting from an initial partition, we iteratively swap vertices across partitions to reduce the cut value while respecting balance constraints.

The planned implementation will be in Python, using standard scientific computing tools such as NumPy, SciPy, and NetworkX. The planned evaluation setup will include grid graphs, Erdos–Renyi random graphs, random geometric graphs, and possibly one small real network. We plan to analyze each method's time complexity and compare cut size, balance ratio, runtime, and scalability as $|V|$ and $|E|$ increase. As an optional application-level experiment, we plan to simulate consensus averaging or distributed gradient descent and measure objective gap, consensus error, and communication cost under different partitions.

5 Project Scope and Expected Goals

The minimum goal is to reproduce the Kernighan–Lin heuristic, implement spectral bisection and a greedy baseline, and compare their partition quality and runtime. This directly demonstrates how classical graph algorithms can address a modern distributed optimization bottleneck: communication-efficient work allocation.

If time permits, we will add one or more extensions:

- weighted graph partitioning with nonuniform communication costs;
- comparison with an existing library such as NetworkX or METIS;
- a stronger distributed optimization simulation with different network topologies and communication budgets.

6 Team Information

Member	Responsibility	Expected Contribution
Name 1	Algorithm implementation	Spectral partitioning and experiments
Name 2	Theory and writing	Problem formulation and complexity analysis
Name 3	Evaluation	Distributed optimization simulation and plots